

RESPONSE TO THE REFEREE:
NEURAL BELLMAN OPERATORS, REVISION R14

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This response addresses the owner-commissioned advisory report dated September 19, 2026, on frozen R13 science commit `dd9e0c7`. The report is not an official Econometrica editorial decision. Each principal comment is mapped to the revised manuscript, proof supplement, and executed evidence. The original theory and historical experiments are preserved without changing their target identities.

1. OVERVIEW OF THE REVISION

We thank the referee for distinguishing the repairs accomplished in the certified response from the requirements of a complete paper. The report has led to a new main manuscript, a current proof supplement, and a unified evidentiary package. The central question is now stated as follows: what can a purchaser establish about a dynamic contract when value information, rather than a selected response, is certified? The finite-query response theorem supplies the information set; the procurement theorem prices its adverse response; the continuous-fee theorem extends the comparison to instruments not evaluated individually.

The revision retains the Neural Bellman Operator framework and the original economic model. It also retains every preceding main-paper and supplemental section in a separately identified historical compendium. Preservation does not turn an old target into current evidence. The new paper states the target, response tolerance, institution, and error interpretation next to each current result. It neither treats the referee's advisory report as an official journal decision nor makes a new editorial recommendation on the referee's behalf.

The most consequential new calculation is the continuous enforcement comparison within the same stochastic settlement economy. The original six-fee menu yields compulsory terms three and four. The 0.05 fee menu yields two and four. A rational interval cover of every fee in $[0, 1]$, both financial mandates, and all eight terms identifies term one in both regimes. The executable adjusted contract requires a smaller surrender charge. Thus the extension does not mechanically reproduce the original duration comparison. Both the original theorem and the stronger continuous-instrument test are presented in the main paper. The result supplies a substantive institutional conclusion: a term comparison made on a coarse enforcement menu can be replaced by a fee or capacity comparison when the enforcement instrument is continuous.

The new witness also applies the multidimensional finite-query theorem to duration and surrender jointly, executes a 54-point opportunity-exposure map including adverse cases, and directly audits the preference variance of the stored constructor. The completed post-review R13 extensions supply the refined menus, actual proposal training, and matched nonneural controls. These are identified as post-review evidence, not as files supposedly available to the earlier referee.

2. RESPONSE TO THE PRINCIPAL COMMENTS

2.1. *T1: A complete paper rather than a code-only response*

The new review object contains `ECTA_R14.pdf`, `SUPP_R14.pdf`, and this response, with editable sources. The main paper is newly written around one theorem chain, rather than a prefatory note attached to an unchanged R12 manuscript. The proof supplement explains the mathematical meaning of the current arithmetic, initial-mandate, oracle, and interval witnesses. The top-level reading index and README identify the new entry point and explicitly label `root ECTA.tex` as historical. The historical R12 main paper, compendium, and supplement are retained unchanged and linked from the current package.

The Econometric Society's `econsocart` class is used without modifying its class or configuration files. The main paper has a conventional abstract, economic environment, principal results, economic application, mechanism and computational sections, conclusion, and author-year bibliography. Repository and workflow details are placed in the supplement and replication notes. The existing author identity and acknowledgments are preserved.

2.2. *T2: Integrating arithmetic and open-mandate semantics*

Section 7.3 of the main paper and Section S.20 of the supplement explain the actual operation model and error recurrence. Binary64 array entries define exact rational data for the stored mathematical problem. The derived all-row maximum for the inherited operations is below 7.057×10^{-11} ; the older records retain their 10^{-7} allowance. New value queries use 10^{-10} only after the independent derivation checks its adequacy. Support calculations have their own error account. This is not a claim about arbitrary sparse-library implementations.

The first positive mandate is the open interval $(0, L]$, not $[0, L]$. A closure optimum at zero supplies an upper supremum only. A feasible positive lower witness is constructed on the adjacent affine financial segment and charged its loss. Exact-response procurement additionally requires attainment. The new executable continuum contracts have strictly positive first actions and independently checked separation from the zero boundary. The original closure/interpolation construction is retained, not replaced by an assertion that zero is positive. Exact-response graph bounds and η -response secants are explicitly separated in Sections 3 and 5.

2.3. *T3: Executed extensions rather than planned evidence*

The report correctly described the extension status at its frozen science commit. The extension execution completed later. We import its immutable deposit separately and identify the source and run in the provenance record. The revision rechecks its 336 dynamic problems, response supports, refined choices, institutional comparisons, rational oracle examples, and primitive examples. The neural training times remain measurements of that completed execution and are labeled accordingly; they are not silently described as repeated timing estimates from the R14 check.

The new continuous-contract, joint-query, and mechanism outputs are generated before release, checked by a separate recursion, and deposited with their arrays and rational inequalities. The referee-facing branch is frozen only after these objects and the manuscripts exist. A source module or a queued workflow is not used as a premise of a numerical statement.

2.4. T4: One target-to-manuscript provenance chain

The current release manifest binds the reviewed report, canonical target manifest, imported extension origin, readable checking source, scientific records, witnesses, generated tables, manuscripts, PDFs, and validation receipts. The mathematical target remains the canonical manifest beginning 54adb353c5b85210d; its full digest appears in the manifest. Every generated current table carries this target identifier and states its error and behavioral conventions.

A source commit identifies the code supplied to execution. A later deposit commit identifies the resulting files. These identities are recorded separately; no self-referential commit identifier is embedded into the file whose commit is being computed. A check-only command hashes the existing inputs, independently recomputes their mathematical content, checks that generated table text agrees with those inputs, and compiles the paper in a temporary directory. It does not regenerate the target or replace the scientific witness. Reproduction of new witnesses is a separate, explicit operation. The distinction is important even when all checks pass.

2.5. E1: Contract refinement in the actual settlement economy

Sections 4 and 5 now extend the actual settlement procurement problem. Proposition 3 separates enforcement capacity E from the surrender charge F . Under the stated technology $F \leq \zeta E$ and a nondecreasing capacity cost that does not change managed wealth, reducing capacity to F/ζ weakly improves purchaser payoff, including when participation is slack. Under binding participation the cost-incidence share cancels from total expenditure. Thus $F = E$ in the baseline follows from the efficient frontier at $\zeta = 1$, rather than from treating the two economic objects as synonyms.

The continuous-fee proof uses policy-value upper inequalities, response supporting planes, exterior surrender bounds, product relaxations for FH , and a cell-wide cost tangent. Exact rational weak duality bounds the purchaser's optimistic payoff throughout each interval. A coverage check excludes gaps. The executable candidate lower bound uses a separately verified all-response duration and a sufficient participation grant. No grid point is treated as an upper bound on the contracts between points.

The resulting compulsory term is one in both regimes. The adjusted executable fee is 0.72880859375, compared with 0.80087890625 without adjustment. The paper reports each global regret upper bound and each competing-term upper bound. It does not claim a unique exact fee optimum or assume attainment of the fee supremum. The original three-versus-four theorem, the refined two-versus-four result, and the continuous one-versus-one result remain side by side. Fee-recipient and quality counterfactuals are explicitly finite-menu calculations unless a continuous cover has also been produced.

2.6. E2: Quantitative model-to-target reasoning

Section 8 and Section S.20 give a model-to-decision transfer theorem. It includes terminal, time-step, truncation, state, action, first-date interpolation, kernel, boundary, and arithmetic entries. The norm, lifting map, and continuation envelope must be specified. Value-query intervals are enlarged by the propagated model error; alternatively a two-sided policy correspondence enlarges behavioral tolerance by 2Δ and charges feature-transfer error separately. The one-dimensional secant calculation shows why the smallest purchaser margin, rather than the scale of private values, controls the acceptable approximation budget.

We also perform a new diagnostic on the actual constructor rather than supplying another hypothetical neighborhood. For the drift-free interior first action, the normalized stored preference variance is approximately 0.0008838834764831865, whereas the antecedent diffusion variance over the same step is 0.0003125. The stored moment is computed as an exact rational function of the array entries and is independently reconstructed. The difference exposes the variance introduced by the state projection. It is not converted into a value-error bound without a regularity argument.

This advances the response by identifying and measuring a concrete approximation mechanism and by proving the full transfer chain. It does not establish quantitative inclusion of the original diffusion inside the purchaser-margin budget. The executed inequalities in the paper concern the specified settlement array economy and continuous contract instruments within it. Continuous-state theory is retained in full; its quantitative economic conclusion requires the missing operator estimates. We have not manufactured those estimates or claimed that a stored-array roundoff certificate supplies them.

2.7. *M1: Neural contribution and end-to-end performance*

Section 7 retains the neural proposal–evaluation–improvement architecture and its theoretical development in the compendium. It now reports actual fitted neural proposals, polynomial controls, and nearest-anchor policies on the same target, with teacher creation, fitting, certified online work, and parameter or policy storage charged separately. The exact-DP upper reference and independent verification are included in the online account.

The recorded neural fit is unfavorable: its largest verified policy gap is about 0.8062, compared with about 0.09017 for the polynomial control and 2.13×10^{-7} for nearest-anchor lookup. The result is retained prominently. It demonstrates that verification is independent of proposal accuracy; it does not demonstrate neural superiority. No claim of high-dimensional speed, repeated-seed reliability, or resource-frontier expansion is inferred from this one target and recorded fit. These empirical claims remain unestablished. The new central response-set and procurement results do not require a neural approximation to be accurate in order to be valid.

2.8. *M2: A central and economically used response-set theorem*

Theorem 1 is now the first principal result. It characterizes the sharp finite-query outer response set through a lifted polyhedron with one candidate response and query-support witnesses. Necessity follows by inserting actual query maximizers. Sufficiency constructs a finite affine-response economy from the proposed witnesses. The scope of sharpness is the class of economies consistent with the oracle information, not the exact attainable-response set of a fixed Markov problem. The proof states its attainment or compact-closure requirement.

Theorem 2 uses supports of this joint set to price participation and purchaser surplus. Theorem 4 uses related query restrictions to cover the continuous enforcement instrument. The actual joint-query experiment evaluates $A + FH$, not a detached artificial statistic. Adding joint directions lowers the positive-mandate upper bound by approximately 0.07348 in the adjusted case, with smaller but positive improvements in the no-adjustment case. Exact and 10^{-4} -response cases are both retained. The experiment and exact rational examples distinguish sharper information from an unsupported assertion that coordinate-wise extremes are jointly feasible.

2.9. M3: Primitive mechanism and adverse parameter map

Section 6 retains the exact opportunity-exposure identity and proves its signed-box enclosure. The identity is presented as a decomposition, not as an economic sign theorem. Proposition 5 adds primitive sufficient conditions in an irreversible-upgrade stopping subclass: an order-preserving wealth transition, nondecreasing consumption bounded by one, common state-independent survival, no separate wealth-dependent surrender option, and an increasing upgrade reward with an upgrade-independent wealth process. Backward induction then proves opportunity-value monotonicity. Stochastic dominance of initial exposure gives the corresponding mandate comparison, with explicit conditions for strictness.

The richer settlement economy is tested separately over 54 combinations of law, operating benefit, fee, and compulsory term. Every positive, negative, and near-zero outcome is retained, together with the local, future, replacement, and total components. Fifteen points have a positive total opportunity difference; the full range includes negative values. The complete continuation arrays and signed-row calculation are independently checked. Neither the primitive subclass nor the grid map is used to assert a universal financial-mandate ranking in the full model.

2.10. C1: A serious nonneural comparison and its scope

The revised computational account includes the standard rectangular parameter-lifting relaxation on the same immutable arrays, with one, two, four, and eight law cells over $[0, 1]$. It is compared with common-law chord and count certificates, with their common feasible lower policies and endpoint work charged. The standard algorithm is attributed to the parametric-verification literature, while the timing is explicitly an in-repository implementation, not an execution of an external package.

The supplement prints the lifting gaps and times, the wider-law results, and the proposal controls. The comparison establishes the behavior of these implementations on this target. It does not establish an external solver ranking, a high-dimensional frontier, or a distribution of random training failures. That distinction is preserved in the main text and in the scope of the validation receipt.

2.11. E3: Economic interpretation and institutional alternatives

Section 2 defines duration as observable continued operation and quality as duration above an observable wealth threshold. The parameter d is a reward-query instrument; the purchaser's service price is a separate parameter. Elective surrender and forced discharge are distinguished, and compulsory terms control the former only. Capacity, fee recipient, outside value, cost incidence, and signing-transfer timing are separately defined.

Section 5 reports alternatives in the same economy: full and half fee receipt by the purchaser; direct and shared capacity costs; capacity efficiency; quadratic and cubic capacity costs; nonlinear compulsory-term costs; quality premia and penalties; outside-value shifts; and service-price shifts. The positive part of the participation grant is retained. One unresolved strict ordering is visibly marked rather than reported as a winner. Binding-participation incidence invariance is proved and tested in the numeraire institution. Moving signing grants or capacity payments into managed wealth is not relabeled as an incidence change; it changes the dynamic problem and is not among the verified interventions. The baseline interpretation is stated so that this limitation is economically explicit.

2.12. *T5 and T6: Arithmetic assumptions and the unit of evidence*

The main paper and supplement describe independent forward-error bounds under an explicit binary64 operation model. They state the maximum sparse row length, discounted mass, value envelope, and operation counts, with separate signed and interpolation charges. They do not call the implementation a formally machine-verified program. Sparse reduction behavior and overflow/underflow assumptions are part of the statement.

The scientific unit is a target, theorem, witness, bound, independent reconstruction, and strict comparison. Workflow success is only the means of producing and checking those objects. Current scientific checks raise explicit failures and their negative fixtures also run under optimized Python. No new output is described as validated until its scientific check has completed.

3. ADDITIONAL TECHNICAL AND ARCHITECTURAL COMMENTS

The main paper concentrates on the response-information–procurement–continuous-instrument chain. The original operator theory, recursive utility, games, regional tools, older experiments, and references are preserved in the complete historical package, with an inventory rather than silent deletion. Current evidence does not depend on reconstructing that chronology.

The exact-response statement precedes the η -robustness discussion. Headline finite-menu tables identify the binding rival. The continuous comparison gives a separate bound for every term, both mandates included; its remaining same-term gap is called regret, not a strict fee-uniqueness margin. The main paper reports the actual $h = 0.0025, 0.005, 0.01$ sensitivity and distinguishes this secant tradeoff from model approximation. Positive lower-witness attainment is checked separately from closure values. All generated current tables carry the same canonical target identity and error convention.

The current README and revision index replace the stale entry point. The original root paper and historical index text remain accessible. The authoritative review command operates on the deposited witness, verifies its immutable inputs, checks the tables, and builds manuscripts separately from scientific generation. Source, result, and deposit identities are deliberately not conflated.

Two substantive empirical requirements are not represented as completed: quantitative diffusion-to-array inclusion at the economic margin, and a demonstrated neural advantage on a scaling frontier with repeated resource-matched controls. The response supplies new theory, actual diagnostics, complete unfavorable results, and testable specifications for those requirements. It does not substitute an acceptance claim, a changed error label, or omitted content for the missing evidence.